

IB MATH AA SL

Lesson 25

Sampling & Describing Data

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Statistics & Probability · 60-minute lesson

Syllabus SL 4.1–4.3 · Property of Lynda Badmus Education

Do Now — recall

ON YOUR WHITEBOARD

1. Find the mean of 4, 6, 8, 10.
2. Find the median of 3, 5, 9.
3. Find the mode of 2, 2, 5, 7.

Answers (click to reveal)

1. 7
2. 5
3. 2

What we're learning today

LEARNING OBJECTIVES

- 1 Identify**
sampling methods and possible bias.
- 2 Calculate**
mean, median and mode.
- 3 Find**
quartiles and the interquartile range.
- 4 Read**
a five-number summary / box plot.

SUCCESS CRITERIA — I CAN...

- ✓
name a sampling technique
- ✓
find the three averages
- ✓
find Q1, Q3 and the IQR
- ✓
describe a data set's centre and spread

Averages & sampling

MEASURES OF CENTRE

$$\bar{x} = \frac{\sum x}{n}$$

Mean

add up, divide by how many

Median

the middle value when ordered

Mode

the most frequent value

SAMPLING

Random

everyone equally likely

Systematic

every kth item

Stratified

by proportion of groups

Watch

bias from a poor sample.

Mean from data

QUESTION

Find the mean of

5, 8, 8, 11, 13.

SOLUTION

Step 1 add the values:

$$\sum x = 5 + 8 + 8 + 11 + 13 = 45$$

Step 2 divide by $n = 5$:

$$\bar{x} = \frac{45}{5} = 9$$

Your turn — averages

ANSWER IN YOUR BOOK

1. Mean of 2, 4, 9.
2. Median of 7, 3, 9, 5.
3. Mode of 1, 4, 4, 4, 9.

Answers (click to reveal)

1. 5
2. 6 (mean of 5 & 7)
3. 4

Spread: quartiles & IQR

FIVE-NUMBER SUMMARY

$$IQR = Q_3 - Q_1$$

Q1

lower quartile (25%)

Q3

upper quartile (75%)

IQR

the middle 50% of the data

BOX PLOT

min, Q1, median, Q3, max

Outlier

$< Q1 - 1.5 \cdot IQR$ or $> Q3 + 1.5 \cdot IQR$

Use

to compare two data sets.

Find the IQR

QUESTION

For the ordered data
3, 5, 7, 8, 12, 14, 18
find Q1, Q3 and the IQR.

SOLUTION

Median is 8. Lower half: 3, 5, 7 $\rightarrow$ Q1 = 5

Upper half: 12, 14, 18 $\rightarrow$ Q3 = 14

IQR = Q3 - Q1:

$$IQR = 14 - 5 = 9$$

Exam-style question

The times (minutes) for 9 runners, in order, are
22, 24, 25, 27, 28, 30, 31, 33, 40.

- (a) Write down the median. [1]
- (b) Find the interquartile range. [3]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) Median = 28 (A1)
- (b) $Q1 = 24.5$, $Q3 = 32$ (M1)(A1) $\rightarrow$ IQR = 7.5 (A1)

Common mistakes



Order first

Always sort the data before finding the median/quartiles.



Mean vs median

An outlier pulls the mean but barely moves the median.



IQR

$IQR = Q3 - Q1$ (a measure of spread, not centre).



Sampling bias

A convenient sample is often not representative.

Mixed practice

QUESTIONS

1. Mean of 10, 20, 30, 40.
2. Median of 2, 8, 4, 6, 10.
3. IQR of 1, 3, 5, 7, 9, 11, 13, 15.
4. Mode of 3, 3, 6, 6, 6, 9.

Answers (click to reveal)

1. 25
2. 6
3. $Q3=12, Q1=4 \rightarrow 8$
4. 6

Plenary & exit ticket

KEY FORMULAS

$$\bar{x} = \frac{\sum x}{n}$$

$$IQR = Q_3 - Q_1$$

EXIT TICKET — before you leave

1. Mean of 6, 6, 12.
2. Median of 4, 9, 2, 7.
3. IQR if $Q_1 = 10$, $Q_3 = 22$.

Answers (click to reveal)

1. 8
2. 5.5
3. 12

Self-assess:
• need more help
• nearly there
• confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Sampling & describing data.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 26

Standard deviation & using your GDC for statistics.

